

Quantum computing

Quantum computing matters because it can tackle problems that are extremely difficult or impossible for today's classical computers

Quantum computing is a method of computation where information is encoded and processed according to the laws of quantum mechanics. The computation is then performed by a quantum device that operates using these quantum principles.

unlike classical computers, which work with bits, quantum computers work with qubits and can use quantum effects such as superposition, entanglement, and interference to solve certain problems more efficiently.

Real-World Problems Quantum Computing May Help Solve

- Healthcare & Drug Discovery – Simulate molecules to develop medicines faster.
- Cryptography & Cybersecurity – Improve encryption and secure communication.
- Logistics & Supply Chain – Optimize routes, deliveries, and resource allocation.
- Finance & Risk Analysis – Enhance portfolio management and fraud detection.
- Climate & Energy Research – Model climate systems and improve renewable energy.
- Quantum Simulations – Study quantum systems that classical computers struggle to model.
- Chemical Simulations – Predict molecular and chemical behavior for research and industry.

Why Quantum Computing Exists ?

As problems become larger and more complex, the number of possible solutions grows rapidly. This requires a huge amount of computation, making some problems difficult for classical computers to solve efficiently.

For example, in drug discovery, simulating the interactions between atoms and molecules becomes increasingly complex as the size of the molecule grows.

Quantum computing uses the principles of quantum mechanics to process information in a different way. Since nature itself follows quantum mechanics, quantum computers can model and solve certain complex problems more naturally and efficiently than classical computers.

Bits: Classical vs Quantum Representation

CLASSICAL BIT

A classical bit can be in **one** of two states:
0 or **1**.



State $|0\rangle$



State $|1\rangle$

Representation

Classical bit is in a definite state:

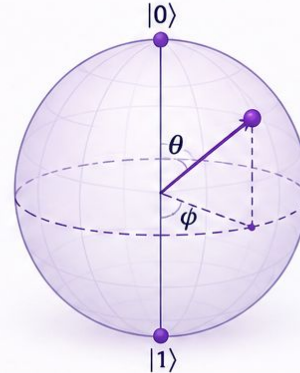
$|0\rangle$ or $|1\rangle$

Key Point

At any given time, a classical bit is either 0 or 1, not both.

QUANTUM BIT (QUBIT)

A qubit can be in a **superposition** of both states simultaneously.



General State

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where $\alpha, \beta \in \mathbb{C}$

$$\text{and } |\alpha|^2 + |\beta|^2 = 1$$

Representation

$$\text{Qubit state } |\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

All points on the Bloch sphere represent valid qubit states.

Key Point

A qubit can be in a superposition of 0 and 1 at the same time.

🎯 States	2 states: $ 0\rangle, 1\rangle$	Infinite states: any superposition $\alpha 0\rangle + \beta 1\rangle$
📊 Measurement	Always gives 0 or 1	Gives 0 with probability $ \alpha ^2$, 1 with probability $ \beta ^2$
📐 Geometry	Two discrete points	All points on the Bloch sphere
💻 Information	1 bit	Up to 1 qubit (can hold more information via superposition)

Superposition

1. What is Superposition?

Superposition is a fundamental property of qubits that allows a qubit to be in multiple states at the same time.

A qubit can be in a combination of $|0\rangle$ and $|1\rangle$ simultaneously.

Mathematically:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α, β are complex numbers such that $|\alpha|^2 + |\beta|^2 = 1$

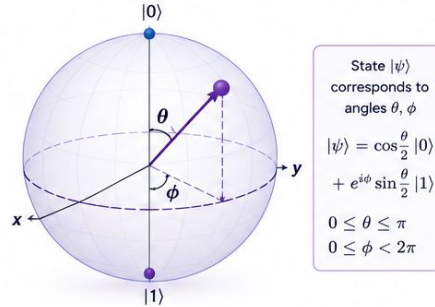
● $|\psi|^2$ = probability of measuring 0

● $|\beta|^2$ = probability of measuring 1

When measured, the qubit “collapses” to 0 with probability $|\alpha|^2$ or to 1 with probability $|\beta|^2$.

2. Visualizing Superposition (Bloch Sphere)

Any qubit state can be represented as a point on the surface of a sphere called the **Bloch sphere**.



State $|\psi\rangle$ corresponds to angles θ, ϕ

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

$$0 \leq \theta \leq \pi$$
$$0 \leq \phi < 2\pi$$

Key Points

- $|0\rangle$ is the north pole ($\theta = 0$). • $|1\rangle$ is the south pole ($\theta = \pi$).
- Points on the equator ($\theta = \pi/2$) are equal superpositions.
- Every point on the surface represents a valid qubit state.

3. Why It Matters

Superposition gives quantum computers their power. It allows them to process many possibilities at once.



Parallelism

A qubit in superposition represents both 0 and 1 at the same time, so n qubits represent 2^n possibilities simultaneously.



More Efficient Computation

Quantum algorithms can evaluate many solutions in parallel, leading to speedups for certain problems.



Interference Enables Solutions

Superposition allows amplitudes to interfere constructively (boosting correct answers) or destructively (canceling wrong ones).



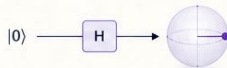
Foundation for Other Quantum Phenomena

Superposition is the basis for entanglement and many other quantum effects.

4. Example: From Superposition to Measurement

Step 1: Create Superposition

Apply a Hadamard gate (H) to $|0\rangle$

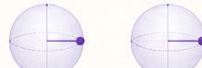


Result: $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Equal probability of 0 or 1 (50% each)

Step 2: Many Qubits

With 2 qubits in superposition:



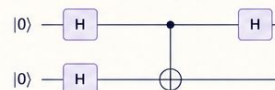
Represents 4 states at once:

$|00\rangle, |01\rangle, |10\rangle, |11\rangle$

(Each with amplitude $1/2$)

Step 3: Quantum Interaction

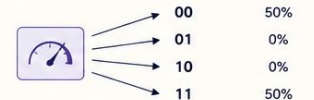
Apply quantum gates



The amplitudes interfere, changing the probabilities of outcomes.

Step 4: Measurement

Measure both qubits



You get one result each time, but over many runs the probabilities match the quantum predictions.

5. Key Takeaways



Superposition means a qubit can be 0 and 1 at the same time with certain probabilities.



It enables quantum computers to explore many possibilities in parallel.



Measurement collapses the state to one outcome, but the probabilities reveal the superposition.



It is a core resource that, together with entanglement and interference, makes quantum computing so powerful.

Entanglement

1. What is Entanglement?

When qubits are entangled, their states are correlated in a way that cannot be explained classically.

Example: Two qubits in an entangled state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$



This means:

- The qubits are not 0 or 1 individually.
- They are linked: when one is measured, the other is determined instantly.

2. How It Works (Example)

Start with two qubits in an entangled state $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$

Measure qubit A	Result	Instant effect on qubit B
A measured 0		B becomes 0
A measured 1		B becomes 1



No signal is sent faster than light.
You only know the correlation after comparing results.

3. Why It Matters



Quantum Cryptography

Entanglement enables unconditional security. If an eavesdropper interferes, the entangled correlations are disturbed and detected.



Quantum Computing

Entangled qubits can represent and process complex relationships, giving quantum computers powerful advantages.



Quantum Communication

Enables future quantum internet and teleportation of quantum states.



Fundamental Insight

Entanglement shows that nature is deeper than classical intuition—information is not just local.

4. Visualizing Entanglement

Classical Correlation (for comparison)

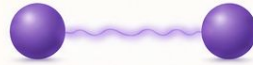
Example: Two coins tied together by a string.



If you know one coin is Heads, you know the other is Tails. But the outcome of each coin was **already determined**.

Entangled Qubits

Two qubits prepared in $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$



The results are **not predetermined**. They are created at the **moment of measurement**, yet always perfectly correlated.

Measurement Outcome



The correlation is perfect, no matter how far apart A and B are.



5. More Entangled States (Examples)

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

Results are opposite (0 ↔ 1).

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

If A is 0, B is 1; if A is 1, B is 0.

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

Opposite results with a phase difference.

6. Key Takeaways



Entanglement links qubits in **non-classical** ways.



It doesn't allow faster-than-light communication.



Measuring one determines the other instantly (with **correlated** results).



It is a key resource for quantum technologies.

Interference

1. What is Interference?

Interference is a quantum phenomenon that occurs when probability amplitudes of a quantum particle combine, causing some outcomes to become more likely and others to become less likely.

Key Idea

A qubit (or quantum particle) can reach the same final state via multiple paths. The amplitudes from those paths add together.

Two types:

- **Constructive interference:** amplitudes add up → higher probability
- **Destructive interference:** amplitudes cancel out → lower (or zero) probability

2. How It Works (Amplitudes Add)

If a quantum system can arrive at the same state via two paths, the total amplitude is the sum of the individual amplitudes.

If the amplitudes are:

In phase (constructive)  Total = A + B (bigger)

Out of phase (by π radians)  Total = A - B (smaller)

Mathematically:

If two paths have amplitudes α and β , the probability of the final state is:

$$P = |\alpha + \beta|^2 = |\alpha|^2 + |\beta|^2 + 2 \operatorname{Re}(\alpha^* \beta)$$

Individual probabilities

Interference term (can be + or -)

3. Why It Matters



Creates Powerful Effects

Interference can boost the probability of correct answers and cancel wrong ones.



Enables Quantum Algorithms

Algorithms like Grover's search and Quantum Fourier Transform use interference to speed up computations.



High Precision

Interference is used in quantum sensors and atomic clocks to achieve extremely high accuracy.

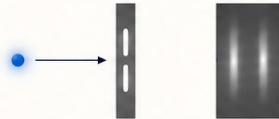


Foundation of Quantum Behavior

It explains patterns in experiments (like the double-slit experiment) that have no classical explanation.

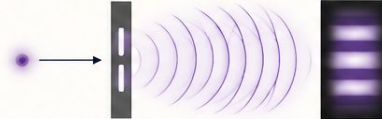
4. Visualizing Interference: The Double-Slit Experiment

Classical Particle (e.g., a ball)



- The particle goes through one slit or the other.
- The pattern on the screen shows two bands.
- No interference (just addition of probabilities).

Quantum Particle (e.g., a photon or electron)



- The particle goes through both slits as a superposition of paths.
- The wave amplitudes from both paths interfere.
- Bright fringes: constructive interference (high probability).
- Dark fringes: destructive interference (zero probability).

How Interference Creates the Pattern

At each point on the screen, the two paths have a phase difference $\Delta\phi$.

Constructive (bright fringe)

$$\Delta\phi = 2\pi n$$

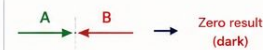
Amplitudes add → high probability



Destructive (dark fringe)

$$\Delta\phi = (2n+1)\pi$$

Amplitudes cancel → zero probability



n is any integer (0, 1, 2, ...).

5. Simple Numerical Example

Suppose two paths have equal amplitudes:

$$\alpha = \frac{1}{\sqrt{2}} e^{i\theta_1}, \quad \beta = \frac{1}{\sqrt{2}} e^{i\theta_2}$$

In phase (constructive)

$$\theta_1 = \theta_2$$

$$P = |\alpha + \beta|^2 = \left| \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right|^2 = 1$$

100% chance of detection at that point.

Out of phase (destructive)

$$\theta_2 = \theta_1 + \pi$$

$$P = |\alpha + \beta|^2 = \left| \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right|^2 = 0$$

0% chance of detection at that point.

6. Key Takeaways

- ✓ Interference happens when probability amplitudes combine.
- ✓ It can increase (constructive) or decrease (destructive) the chance of outcomes.
- ✓ It is a key reason quantum computers can outperform classical computers.
- ✓ Interference is a uniquely quantum effect—no classical analogue exists.

Real-World Analogy

Think of ocean waves: Two waves meeting can make a bigger wave (constructive) or cancel out to make a calm sea (destructive).



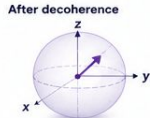
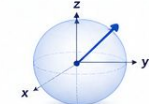
Decoherence

1. What is Decoherence?

Decoherence is the process by which a quantum system loses its quantum behavior due to unwanted interactions with its environment.

It doesn't destroy the system's energy, but it destroys its ability to behave quantum mechanically.

Example: A qubit in superposition



Pure quantum state
 $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
 (Coherent superposition)

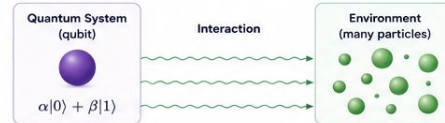
Shows quantum behavior
 (interference, etc.)

Mixed state (classical mixture)
 $\rho = |\alpha|^2|0\rangle\langle 0| + |\beta|^2|1\rangle\langle 1|$
 (No coherence)

No quantum interference;
 behaves classically.

2. How Decoherence Happens

A quantum system interacts with many tiny degrees of freedom in its environment (like air molecules, photons, vibrations, etc.). This leaks "which-path" or phase information to the environment.



As information flows into the environment, the relative phase between quantum states becomes random and uncontrollable.

Mathematical picture (intuition)

The off-diagonal terms (coherences) in the density matrix decay.

$$\rho_{\text{pure}} = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix} \xrightarrow{\text{decoherence}} \rho_{\text{mixed}} = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$$

Coherent terms (enable interference) Coherences vanish (no interference)

3. Why It Matters



Limits Quantum Behavior

Decoherence explains why macroscopic objects don't show quantum effects.



Major Challenge for Quantum Computing

It causes qubits to lose coherence, leading to errors in quantum computations.



Sets Coherence Time

The "coherence time" is how long a qubit can maintain quantum information before decohering.



Drives Need for Error Correction

Quantum error correction and fault-tolerant designs exist to fight decoherence.



Fundamental to Quantum-Classical Transition

Decoherence explains how the classical world emerges from quantum rules.

4. Example: Decoherence Over Time (Qubit)

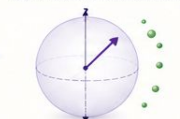
Initial state (pure)
 Qubit in superposition

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$



Short time

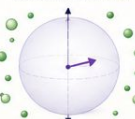
Slight interaction with environment



Some phase information leaked

Longer time

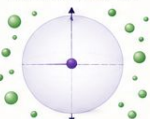
More information leaked



Coherence reduced

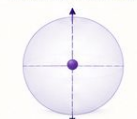
Even longer

Environment knows more



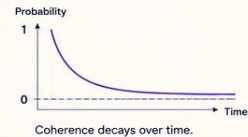
Almost no coherence

After decoherence
 Effectively classical mixture



No preferred phase left

What you observe
 Interference disappears.



5. Real-World Sources of Decoherence



Thermal noise
 (Random molecular collisions)



Electromagnetic fluctuations
 (Stray fields, blackbody radiation)



Vibrations
 (Mechanical coupling)



Acoustic noise
 (Sound waves)



Other qubits
 (Unwanted cross-talk)

Anything that can gain "which-path" or phase information about the system can cause decoherence.

6. Can Decoherence Be Prevented?

Approach	How it Helps
Isolation	Physically isolate qubits from environment.
Low temperature	Reduces thermal noise.
Dynamical decoupling	Uses pulses to average out noise.
Error correction	Detects and corrects errors caused by decoherence.
Better materials & design	Reduces unwanted interactions.

We can't eliminate decoherence completely, but we can slow it down and correct its effects.

7. Key Takeaways

- ✓ Decoherence is the loss of quantum coherence due to environmental interactions.
- ✓ It turns pure quantum states into classical mixtures.
- ✓ It destroys interference, making the system behave classically.
- ✓ It limits the performance of quantum technologies.
- ✓ Understanding and controlling decoherence is essential for building practical quantum devices.

Decoherence is not a bug in quantum mechanics—it's how the quantum world connects to the classical world.

Classical vs Quantum Computing

	1. Basic Unit of Information												
BIT A bit can be either 0 or 1. It exists in one state at a time.													
0 or 1 State: 0 State: 1													
	2. How It Works												
DETERMINISTIC Follows a fixed set of instructions. Same input always gives the same output.													
Input →  Processing (Logic Gates) → Output													
	3. Information Processing												
SEQUENTIAL Processes one possibility at a time. Explores solutions step-by-step.													
 Explores one path at a time.													
	4. Power & Performance												
EFFICIENT FOR - Simple calculations - Everyday tasks - Well-structured problems - Tasks with known algorithms 													
	5. Example Operations												
Addition of 2 numbers (example) $5 (0101) + 3 (0011) = 8 (1000)$ Done using logic gates step-by-step.													
<table><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr></table>		0	1	0	1	0	0	1	1	1	0	0	0
0	1	0	1										
0	0	1	1										
1	0	0	0										
	6. Hardware												
Built with transistors (silicon chips). Stable, works at room temperature. Mature technology.													
													

QUBIT A qubit can be 0, 1, or both at the same time (superposition). It exists in multiple states simultaneously.	
	Superposition $\alpha 0\rangle + \beta 1\rangle$
PROBABILISTIC Uses quantum mechanics. Measurement gives probabilistic outcomes.	
Input →  Quantum Processing → Measurement  → Output (Probabilistic)	
PARALLEL (QUANTUM PARALLELISM) Processes many possibilities at once thanks to superposition. Explores many paths simultaneously.	
	
POWERFUL FOR - Complex problems - Large search spaces - Optimization - Simulation of quantum systems - Problems hard for classical computers 	
Quantum Parallelism (example) A quantum computer with n qubits can represent 2^n states at once. With 3 qubits $\rightarrow 2^3 = 8$ states simultaneously.	
000 001 010 011 100 101 110 111 All at once!	
Built with qubits using technologies like: superconducting circuits, trapped ions, photons, etc. Requires extreme conditions (very low temperature, isolation from noise).	
	

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7. Where They Are Used	
Classical Computers	Quantum Computers (Early Stage)
 Laptops, phones, servers  Web browsing  Office apps  Games  Banking systems  Most of today's software	 Drug discovery  Material science  Financial modeling  Optimization (logistics, scheduling)  Cryptography  AI & machine learning (future)

8. Key Differences at a Glance		
Feature	Classical Computing	Quantum Computing
Basic unit	Bit (0 or 1)	Qubit (0, 1, or both)
State	One state at a time	Many states at once
Processing	Sequential	Parallel (via superposition)
Outcome	Deterministic	Probabilistic
Power	Good for general tasks	Powerful for specific complex tasks
Technology	Mature, stable	New, developing

9. Key Takeaway	
	Classical computers are great for everyday computing.
	Quantum computers are designed to solve certain complex problems that are extremely hard or impossible for classical computers.
	They are not replacements, but complements to each other.

IBM Quantum Learning Ecosystem

Why IBM Matters

IBM helps people learn and use quantum computing by providing:

- **Real Quantum Hardware** – Access to actual quantum computers.
- **Cloud Access** – Run quantum programs online from anywhere.
- **Learning Resources** – Tutorials, courses, and documentation.
- **Development Tools** – Software for building and testing quantum programs.

IBM Quantum Stack

Developer → Qiskit → IBM Runtime → Quantum Hardware → Results

What is Qiskit?

Qiskit is IBM's open-source Software Development Kit (SDK) for quantum computing.

What Can You Do with Qiskit?

- Build quantum circuits
- Simulate quantum programs
- Run programs on real IBM quantum computers
- Analyze and visualize results

Key Takeaway

Qiskit allows developers and learners to create, test, and run quantum applications using IBM's quantum computing platform.